1 Rearranges to $5y = -2x + 15$ to find gradient of L is $-\frac{2}{5}$ **[M1]**

$\frac{5}{2}$ or 2.5 **[A1]**

[Total Marks: 2]

2 Identifies the gradient of the given line as 3 **[B1]**

Uses the perpendicular gradient $-\frac{1}{3}$ and substitutes $(6, 2)$ into $y = mx + c$ or $y - y_1 = m(x - x_1)$ **[M1]**

$-\frac{1}{3}x + 4$ **[A1]**

[Total Marks: 3]

3 (a) -2 **[B1]**

(b) Gradient of $L_2 = \frac{1}{2}$ **[M1]**

$1 = \frac{1}{2}(4) + c$ or $y - 1 = \frac{1}{2}(x - 4)$ **[M1]**

$y = \frac{1}{2}x - 1$ **[A1]**

[Total Marks: 4]

4 (a) Finds gradient of L_1: $y = \frac{3}{4}x - 2$ giving $m_1 = \frac{3}{4}$ **[M1]**

$-\frac{4}{3}$ **[A1]**

(b) Substitutes $(6, -5)$ and $m = -\frac{4}{3}$ into $y = mx + c$: $-5 = -\frac{4}{3}(6) + c$ **[M1]**

$-\frac{4}{3}x + 3$ **[A1]**

[Total Marks: 4]

5 (a) 2 **[B1]**

(b) $-\frac{1}{2}$ (or -0.5) **[B1]**

(c) $3 = -\frac{1}{2}(4) + c$ or $y - 3 = -\frac{1}{2}(x - 4)$ **[M1]**

$c = 5$ **[A1]**

$-\frac{1}{2}x + 5$ **[A1]**

[Total Marks: 5]

6 **(a)** Rearranges the equation to make y the subject: $y = \frac{5}{2}x + 4$ **[M1]**

$\frac{5}{2}$ or 2.5 **[A1]**

(b) $-\frac{2}{5}$ or -0.4 **[B1]**

(c) Substitutes $(10, -3)$ and $m = -\frac{2}{5}$ into linear equation: $-3 = -\frac{2}{5}(10) + c$ or $y - (-3) = -\frac{2}{5}(x - 10)$ **[M1]**

Solves for c: $-3 = -4 + c \implies c = 1$ **[M1]**

$-\frac{2}{5}x + 1$ (or $-0.4x + 1$) **[A1]**

[Total Marks: 6]

7 **(a)** Midpoint $= \left(\frac{-3+9}{2}, \frac{8+(-2)}{2}\right) = (3, 3)$ **[B1]**

Gradient $= \frac{-2-8}{9-(-3)} = -\frac{10}{12} = -\frac{5}{6}$ **[B1]**

(b) Uses perpendicular gradient $m_\perp = \frac{6}{5}$ **[M1]**

Substitutes midpoint $(3, 3)$: $3 = \frac{6}{5}(3) + c$ or $y - 3 = \frac{6}{5}(x - 3)$ **[M1]**

$6x - 5y = 3$ (or $-6x + 5y = -3$) **[A1]**

[Total Marks: 5]

8 Calculates the midpoint of $AB = \left(\frac{-4+2}{2}, \frac{7+(-1)}{2}\right) = (-1, 3)$ **[B1]**

Calculates the gradient of $AB = \frac{-1-7}{2-(-4)} = -\frac{8}{6} = -\frac{4}{3}$ **[B1]**

Finds the perpendicular gradient $= \frac{3}{4}$ **[M1]**

Substitutes the midpoint $(-1, 3)$ and gradient $\frac{3}{4}$ to find the equation, e.g., $y - 3 = \frac{3}{4}(x - (-1))$ **[M1]**

$3x - 4y = -15$ or $4y - 3x = 15$ or equivalent integer form **[A1]**

[Total Marks: 5]

9 **(a)** Gradient of $L = -\frac{2}{3}$ **[M1]**

$\frac{3}{2}$ (or 1.5) **[A1]**

(b) **(i)** $\left(\frac{-3+9}{2}, \frac{8+(-2)}{2}\right)$ **[M1]**

$(3, 3)$ **[A1]**

(ii) Gradient of $AB = \dfrac{-2-8}{9-(-3)} = -\dfrac{5}{6}$ **[M1]**

Gradient of perpendicular bisector $= \dfrac{6}{5}$ **[A1]**

$y - 3 = \dfrac{6}{5}(x-3)$ **[M1]**

$6x - 5y = 3$ (or $-6x + 5y = -3$) **[A1]**

[Total Marks: 8]

10 (a) $\left(\dfrac{-3+9}{2}, \dfrac{8+(-2)}{2}\right)$ **[M1]**

$(3, 3)$ **[A1]**

(b) Gradient of $AB = \dfrac{-2-8}{9-(-3)}$ or $-\dfrac{10}{12}$ or $-\dfrac{5}{6}$ **[M1]**

Perpendicular gradient is $\dfrac{6}{5}$ **[M1]**

Substitute $(3, 3)$ into $y - 3 = \dfrac{6}{5}(x-3)$ or $3 = \dfrac{6}{5}(3) + c$ **[M1]**

$6x - 5y = 3$ or $-6x + 5y = -3$ **[A1]**

[Total Marks: 6]

11 (a) $(0, 6)$ **[B1]**

(b) Gradient of $L_1 = -\dfrac{3}{2}$ **[M1]**

Gradient of $L_2 = \dfrac{2}{3}$ **[M1]**

$y = \dfrac{2}{3}x + 6$ **[A1]**

(c) Coordinates of Q: $y = \dfrac{2}{3}(9) + 6 = 12 \implies Q(9, 12)$ **[B1]**

Coordinates of R: $3(4) + 2y = 12 \implies y = 0 \implies R(4, 0)$ **[B1]**

Identify triangle PQR is right-angled at P OR setup valid box/shoelace method **[M1]**

Length $PQ = \sqrt{9^2 + 6^2} = \sqrt{117}$ and Length $PR = \sqrt{4^2 + (-6)^2} = \sqrt{52}$ **[M1]**

Area $= \dfrac{1}{2} \times \sqrt{117} \times \sqrt{52} = 39$ **[A1]**

[Total Marks: 9]

12 (a) Rearranges to $2y = -kx + 12$ to identify gradient $-\dfrac{k}{2}$ **[B1]**

(b) Uses the gradient formula: $\dfrac{(k+2)-5}{2-0}$ **[M1]**

$\dfrac{k-3}{2}$ **[A1]**

(c) Applies the perpendicularity condition $m_1 \times m_2 = -1$ **[M1]**

Forms the equation: $\left(-\dfrac{k}{2}\right)\left(\dfrac{k-3}{2}\right) = -1$ **[M1]**

Expands and rearranges to form a quadratic equation:
$-k(k-3) = -4 \implies k^2 - 3k - 4 = 0$ **[M1]**

4 and -1 **[A1]**

(d) Deduces that $k = -1$ must be chosen because $m_1 = -\dfrac{k}{2}$ must be positive (yielding $m_1 = \dfrac{1}{2}$) **[M1]**

Calculates the gradient of L_2 when $k = -1$: $m_2 = \dfrac{-1-3}{2} = -2$, and uses the y-intercept from point $A(0, 5)$ to write $y = -2x + 5$ **[M1]**

$2x + y = 5$ **[A1]**

[Total Marks: 10]

13 $4y = 3x - 12$ implies gradient of P is $\dfrac{3}{4}$ **[M1]**

Gradient of Q is $-\dfrac{4}{3}$ **[M1]**

Substitute $(6, -2)$ into $y = -\dfrac{4}{3}x + c$ or $y - (-2) = -\dfrac{4}{3}(x - 6)$ **[M1]**

$-\dfrac{4}{3}x + 6$ **[A1]**

[Total Marks: 4]

14 (a) $\left(\dfrac{-2+6}{2}, \dfrac{4+0}{2}\right)$ **[M1]**

$(2, 2)$ **[A1]**

(b) Gradient of $AB = \dfrac{0-4}{6-(-2)} = -\dfrac{1}{2}$ **[M1]**

Gradient of perpendicular bisector $= 2$ **[M1]**

Substitute midpoint $(2, 2)$ into $y - y_1 = m(x - x_1)$: $y - 2 = 2(x - 2)$ **[M1]**

$y = 2x - 2$ **[A1]**

[Total Marks: 6]

15 (a) Uses the midpoint formula: $\left(\dfrac{-4+2}{2}, \dfrac{5+(-7)}{2}\right)$ **[M1]**

$(-1, -1)$ **[A1]**

(b) Uses the gradient formula: $\dfrac{-7-5}{2-(-4)}$ or $\dfrac{5-(-7)}{-4-2}$ **[M1]**

$\frac{-12}{6} = -2$ **[A1]**

(c) Identifies the perpendicular gradient is $\frac{1}{2}$ (or 0.5) **[B1]**

Substitutes their midpoint $(-1, -1)$ and perpendicular gradient into $y = mx + c$:
$-1 = \frac{1}{2}(-1) + c$ **[M1]**

$\frac{1}{2}x - \frac{1}{2}$ (or $0.5x - 0.5$) **[A1]**

(d) Sets $y = 0$ in their equation from part (c): $0 = \frac{1}{2}x - \frac{1}{2}$ **[M1]**

$(1, 0)$ **[A1]**

[Total Marks: 9]

16 (a) Calculates gradient of AB: $\frac{9-1}{2-(-2)} = 2$ **[B1]**

Uses perpendicular gradient $m_{BC} = -\frac{1}{2}$ (or -0.5) **[M1]**

Substitutes $B(2, 9)$: $9 = -\frac{1}{2}(2) + c$, giving $y = -\frac{1}{2}x + 10$ (or $-0.5x + 10$) **[A1]**

(b) Substitutes $y = 0$ into equation of BC: $0 = -0.5x + 10$ **[M1]**

$(20, 0)$ **[A1]**

(c) Finds $AB = \sqrt{4^2 + 8^2} = \sqrt{80}$ and $BC = \sqrt{18^2 + (-9)^2} = \sqrt{405}$ **[M1]**

$\frac{1}{2} \times \sqrt{80} \times \sqrt{405} = 90$ **[A1]**

[Total Marks: 7]

17 (a) Substitutes $x = 0$ into $2y + 3x = 12$ because P is on the y-axis **[M1]**

$(0, 6)$ **[A1]**

(b) Rearranges L_1 equation to find gradient: $2y = -3x + 12 \implies y = -\frac{3}{2}x + 6$, so $m = -\frac{3}{2}$ **[B1]**

States perpendicular gradient for L_2 is $\frac{2}{3}$ **[M1]**

$\frac{2}{3}x + 6$ **[A1]**

(c) Substitutes $y = 0$ into the equation of L_2 to find the coordinates of Q **[M1]**

$0 = \frac{2}{3}x + 6 \implies x = -9$ **[A1]**

Uses the distance formula for $P(0, 6)$ and $Q(-9, 0)$: $\sqrt{(-9-0)^2 + (0-6)^2}$ **[M1]**

$\sqrt{117}$ or $3\sqrt{13}$ **[A1]**

[Total Marks: 9]

18 **(a)** Gradient of $AB = \dfrac{5-2}{5-(-1)} = \dfrac{1}{2}$ **[M1]**

Gradient of $BC = -2$ **[A1]**

$-2x + 15$ **[A1]**

(b) $0 = -2x + 15$ **[M1]**

$(7.5, 0)$ **[A1]**

(c) $AB = \sqrt{(5-(-1))^2 + (5-2)^2} = \sqrt{45}$ and $BC = \sqrt{(7.5-5)^2 + (0-5)^2} = \sqrt{31.25}$ **[M1]**

Area $= \dfrac{1}{2} \times \sqrt{45} \times \sqrt{31.25}$ **[M1]**

18.75 **[A1]**

(d) Midpoint of $AC = \left(\dfrac{-1+7.5}{2}, \dfrac{2+0}{2}\right) = (3.25, 1)$ **[B1]**

Gradient of $AC = \dfrac{0-2}{7.5-(-1)} = -\dfrac{4}{17}$, so perpendicular gradient $= \dfrac{17}{4}$ $(= 4.25)$ **[M1]**

$y - 1 = 4.25(0 - 3.25)$ **[M1]**

$(0, -12.8)$ **[A1]**

[Total Marks: 12]